

Yuan Pu

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Summer 2027 availability: May 1 – August 31

EDUCATION

Duke University

Ph.D. in Computer Science

August 2025 – Present (expected graduation: before May, 2030)

Durham, NC

- GPA: 4.0/4.0

Brown University

Bachelor of Science in Computational Biology (Computer Science track)

August 2019 – May 2023

Providence, RI

- GPA: 4.0/4.0; magna cum laude; Computational Biology Departmental Honors

RESEARCH & TECHNICAL SKILLS

Research Areas: Natural Language Processing, Evaluation, Human-AI Interaction, LLM Safety & Reliability, AI for Health

ML Methods: Generative Modeling, Statistical Modeling, Deep Learning, Time-Series Modeling

Programming Languages: Python, R, MATLAB

Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, NumPy, pandas, Stan

Tools & Platforms: Git, Jupyter, OpenAI/Anthropic APIs, Weights & Biases, Google Cloud Platform, HPC clusters

Open Source: [MedRedFlag](#) (code/data), [Trajectory Flow Matching](#) (code)

SELECTED PUBLICATIONS & PREPRINTS (*CO-FIRST AUTHORSHIP)

MedRedFlag: Investigating how LLMs Redirect Misconceptions in Real-World Health Communication | *ACL 2026 (Findings), SAIL 2026*

Sraavya Sambara*, [Yuan Pu](#)*, Ayman Ali, Vishala Mishra, Lionel Wong, Monica Agrawal

Counting Clues: A Lightweight Probabilistic Baseline Can Match an LLM | *AACL-IJCNLP 2026 (Main); ML4H 2025 (Findings)*

Furong Jia*, [Yuan Pu](#)*, Finn Guo, Monica Agrawal

Trajectory Flow Matching with Applications to Clinical Time Series Modelling | *NeurIPS 2024 (Spotlight)*

Xi Zhang*, [Yuan Pu](#)*, Yuki Kawamura, Andrew Loza, Yoshua Bengio, Dennis Shung, Alexander Tong

Human-Algorithmic Interaction Using a Large Language Model-Augmented Artificial Intelligence Clinical Decision Support System | *CHI 2024*

Niroop Channa Rajashekar*, Yeo Eun Shin*, [Yuan Pu](#)*, et al.

Challenges of Auditing: Variability in Outputs of Large Language Models for Health | *preprint, 2026*

[Yuan Pu](#), Yewon Chang, Furong Jia, Xunjian Yin, Jessica Ma, Ayman Ali, Monica Agrawal

RESEARCH EXPERIENCE

Graduate Student

Duke University

Aug 2025 – Present

Durham, NC

User-Centered Evaluation of LLMs in Real-World Use | *advised by Monica Agrawal*

- Co-developed [MedRedFlag](#), a benchmark of 1,100+ real-world patient questions with problematic assumptions; built a semi-automated LLM annotation pipeline and LLM-as-judge metrics, and benchmarked frontier LLMs against physician responses, revealing systematic gaps in misconception handling. (ACL 2026 Findings; SAIL 2026, co-first author)
- Led a controlled audit of frontier LLMs across API and consumer-facing access modes; collected repeated responses to real-world health questions and characterized differences in response form, interaction behavior, referenced sources, and clinical content, finding systematic cross-mode differences that limit API-only evaluations. (preprint, first author)

Probing LLM Reasoning with Probabilistic Baselines | *advised by Monica Agrawal*

- Co-designed a lightweight Naive Bayes method over concept-diagnosis co-occurrence statistics from LLM pretraining corpora. Demonstrated that FBPR matches 7B LLM accuracy on MedQA diagnosis benchmarks with complementary error patterns, probing the role of corpus frequency in LLM clinical reasoning. (AACL-IJCNLP 2026 Main, accepted; ML4H 2025 Findings, co-first author)

Postgraduate Associate

Yale School of Medicine

Aug 2023 – May 2025

New Haven, CT

Generative Modeling for Time Series with Flow Matching | *advised by Dennis Shung and Alex Tong*

- Co-developed Trajectory Flow Matching in PyTorch, a scalable generative framework for training Neural SDEs on irregularly sampled time series with uncertainty quantification, interval prediction, and conditional generation; co-led implementation of the model and training pipeline, and ran large-scale experiments. (NeurIPS 2024 Spotlight, co-first author)

Evaluating Usability and Trust in an LLM-Augmented Decision Support System | *advised by Dennis Shung*

- Co-led a mixed-methods evaluation of an LLM-augmented clinical decision support system, measuring usability, trust calibration, and AI-assisted decision-making; facilitated medical simulations with 106 clinicians and led quantitative survey analysis and thematic coding of interviews. (CHI 2024, co-first author)

Large-Scale Clinical Cohort Data Analysis | *advised by Dennis Shung and Darrick Li*

- Engineered a data extraction and analysis pipeline for a 670K-patient clinical database; applied survival analysis and risk-factor characterization to post-PCI GI bleeding. (Alimentary Pharmacology & Therapeutics 2026, co-first author)

Undergraduate Research Assistant

Brown University

Sept 2020 – May 2023

Providence, RI

Deep Learning for Gene Expression Prediction from Epigenetic Data | *advised by Ritambhara Singh*

- Built attention-augmented RNNs in Python to predict gene expression from multimodal epigenetic features across patients; conducted cross-patient transfer analysis on glioblastoma stem-cell data to identify predictive features. (PLOS computational biology 2025)

Biomarker Discovery in Cancer Genomics | *advised by Mamiko Yajima*

- Analyzed clinical records and genomic profiles of AML patients, applying survival analysis and differential expression methods to identify prognostic biomarkers and enriched pathways.

Data Science Intern

Brown University Computational Biology Core

Jan 2022 – May 2023

Providence, RI

Small RNA-seq Data Pipeline for Toxicogenomics | *advised by Daniel Spade and August Guang*

- Built a small RNA-seq pipeline to identify dose-dependent expression changes across conditions, spanning raw sequencing data processing through pathway-level biological interpretation. (Toxicological Sciences 2026, first author)

ADDITIONAL EXPERIENCE & HONORS

Reviewer: ML4H, CSCW, HAI, COLM DAIH	2025–2026
Roundtable Junior Chair, Clinician-AI Interaction ML4H	2024
Research Intern, AI-powered 3D interior design Dymaxion	Nov 2023–Aug 2024
Duke University Summer Research Fellowship;	2026
SAIL Travel Support	2026
NeurIPS Scholar Award	2024
Sigma Xi Scientific Honor Society	2023

TEACHING EXPERIENCE

Duke University: COMPSCI375/575 Natural Language Processing (Bhuwan Dhingra)	Fall 2026
Brown University: CSCI1430 Computer Vision (James Tompkin)	Spring 2022, 2023
Brown University: CSCI1810/2810 Computational Molecular Biology (Sorin Istrail)	Fall 2022